

Proof-Carrying Finite Planar Three-Body Continuation with Repeated Levi–Civita Chart Switching

Jeffrey Spaulding
Independent Researcher
jspaulding42@hotmail.com

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Abstract

We present an implemented, proof-carrying continuation checker for one supplied planar Newtonian three-body point initial-value problem. Every accepted raw real numeric field is a finite binary64 value interpreted, in theorem-facing exact calculations, as its exact dyadic rational. A raw version-1 certificate has the finite grammar

$$N(N \rightarrow N \mid N \rightarrow \text{LC}_{ij} \rightarrow N)^*, \quad (i, j) \in \{(0, 1), (0, 2), (1, 2)\},$$

so all three canonical binary pairs may occur and a pair may be revisited after returning to ordinary coordinates. The version-2 replay checker revalidates every ordinary and Levi–Civita (LC) tube, complete handoff box, local deck-gauge relation, and conditional physical-clock update. LC mass coefficients on this theorem surface are derived from exact mass fractions and enclosed outward in binary64 arithmetic. A successful finite induction produces an exact-clock preimage of a requested physical time and an exact `RationalInterval` Horner enclosure of the state there. The two mathematical outcomes are `CERTIFIED_TO_T` and a structured `UNRESOLVED` frontier; malformed wire data may instead fail before replay. The result is replayed by the same implementation and is not an independent formal verification. We also retain a separate, stronger local theorem for a chart with an exact binary-collision event anchor. An ordinary LC passage does not by itself prove that a collision occurs. The contribution is a sound finite continuation kernel for a specified exact planar IVP, not a complete producer, a closed form, or an all-time solution of the three-body problem.

1 Introduction

The Newtonian three-body flow is classical away from collision, but direct Cartesian coordinates become singular when two bodies meet. In the planar problem the Levi–Civita transformation replaces a selected binary collision by an analytic lifted differential equation [3, 8, 5]. Regularization alone, however, does not show that a computed trajectory encloses an exact solution, that successive charts describe the same initial-value problem, or that their local time coordinates form one physical clock. These are proof obligations, not matters of plotting accuracy.

This paper describes an executable acceptance theorem for a *supplied finite chain*. Candidate generation is deliberately outside the theorem. The checker starts with one exact serialized point IVP, freshly validates the initial ordinary Newtonian chart, and folds a strict sequence of ordinary bridges and complete ordinary–LC–ordinary passages. At every accepted vertex it carries one actual solution branch and an interval containing that branch’s actual clock origin. It never promotes all points of a rectangular tube or all clocks in an interval to realizable trajectories. Complete box containment is used to retain the one actual state despite discarded correlations.

The main contributions are:

- (i) a strict raw certificate grammar whose records do not contain trusted intermediate checker results, propagated clocks, gauge choices, or final enclosures;
- (ii) a finite conditional-induction theorem for ordinary bridges and for LC passages of any canonical binary pair, including repeated visits;
- (iii) exact forward clock-ledger rules and exact fixed-physical-time evaluation after the final ordinary chart;
- (iv) an outward mass-coefficient kernel which supports generic positive binary64 mass triples on the new LC-tube/carried-entry/carried-exit surface without requiring every derived ratio to be exactly binary64; and
- (v) fail-closed replay with either a certified terminal enclosure or a typed unresolved frontier retaining only the proved prefix.

The design follows the established validated-integration viewpoint [4, 7, 6], but it focuses on composition: exact IVP identity, complete transition containment, chart gauge, and the physical-time cocycle are part of the checked artifact. Capiński, Kepley, and Mireles James proved substantially stronger orbit-level collision and near-collision conclusions in the circular restricted three-body problem using computer-assisted methods [1]. That is a different dynamical model and proof goal. The present result concerns one supplied finite chain in the full planar three-body problem and remains conditional on the trusted kernel listed in Section 10. No novelty or priority relative to that work is asserted.

The term “computational solution” is used here in a deliberately limited sense: if a finite certificate is accepted, the checker encloses the exact state at one requested finite time. A general certified computational solution would additionally need a producer and a theorem establishing coverage, progress, and suitable termination for a broad input class. Those properties are not consequences of the acceptance theorem proved here.

2 Ordinary and regularized dynamics

2.1 Planar Newtonian charts

Let the positive masses be m_0, m_1, m_2 . For positions $q_a \in \mathbb{R}^2$ and velocities $v_a \in \mathbb{R}^2$, $a \in \{0, 1, 2\}$, the ordinary state $x = (q_0, q_1, q_2, v_0, v_1, v_2) \in \mathbb{R}^{12}$ obeys

$$\dot{q}_a = v_a, \quad \dot{v}_a = \sum_{b \neq a} m_b \frac{q_b - q_a}{|q_b - q_a|^3}. \quad (1)$$

An ordinary chart uses a local parameter s . Along the carried physical solution its physical time has the affine form

$$t = s + b, \quad (2)$$

where b is the chart’s clock origin. The field is analytic and locally unique on any box for which every pair distance has a positive lower bound.

2.2 A selected Levi–Civita chart

Choose a canonical ordered pair (i, j) , with $i < j$, and let k be the remaining body. Put

$$M = m_i + m_j, \quad \alpha = \frac{m_j}{M}, \quad \beta = \frac{m_i}{M}.$$

For $z = (z_1, z_2)$ define

$$Q(z) = (z_1^2 - z_2^2, 2z_1 z_2), \quad \rho = |z|^2, \quad L(z) = 2 \begin{pmatrix} z_1 & -z_2 \\ z_2 & z_1 \end{pmatrix}.$$

The lifted state is

$$X = (z, w, h, R, U, y, V, t) \in \mathbb{R}^{14}.$$

Here $q = Q(z)$ is the selected relative position, R is its pair center of mass, and y is the displacement of the third body from R . Define

$$\begin{aligned} d_i &= y + \alpha Q(z), & d_j &= y - \beta Q(z), & f_i &= \frac{d_i}{|d_i|^3}, & f_j &= \frac{d_j}{|d_j|^3}, \\ B &= \frac{m_k}{M}(m_i f_i + m_j f_j), & A_k &= -m_i f_i - m_j f_j, \\ Y &= A_k - B, & P &= m_k(f_j - f_i). \end{aligned}$$

The regularized field used by the checker is

$$\begin{aligned} z' &= w, & w' &= \frac{1}{2}hz + \frac{1}{4}\rho L(z)^T P, \\ h' &= (L(z)w) \cdot P, & R' &= \rho U, & U' &= \rho B, \\ y' &= \rho V, & V' &= \rho Y, & t' &= \rho. \end{aligned} \tag{3}$$

Primes denote the LC parameter. There is no division by z or by ρ ; the field is analytic at a selected binary collision while d_i and d_j remain separated from zero.

On a punctured LC interval, the physical relative velocity is

$$v = \frac{L(z)w}{\rho}. \tag{4}$$

The Cartesian reconstruction is

$$q_i = R - \alpha q, \quad q_j = R + \beta q, \quad q_k = R + y,$$

with the analogous velocity formulas using U , v , and V . The deck action

$$\Gamma(z, w, h, R, U, y, V, t) = (-z, -w, h, R, U, y, V, t) \tag{5}$$

does not change the physical projection.

The pair-energy constraint is

$$C = 2|w|^2 - M - \rho h. \tag{6}$$

Lemma 2.1 (Constraint and punctured projection). *For the exact field (3), $C' = 0$. If $C = 0$ and $\rho > 0$, the projection (4) with $dt/ds = \rho$ satisfies the Newtonian equations (1). Moreover $F(\Gamma X) = D\Gamma F(X)$.*

Proof. Since $\rho' = 2z \cdot w$, substitution in (3) gives

$$C' = 4w \cdot w' - \rho' h - \rho h' = 0.$$

The identities

$$L^T L = L L^T = 4\rho I, \quad Lz = 2Q(z),$$

and

$$\rho L(w)w - \rho' L(z)w = -2|w|^2 Q(z)$$

reduce the projected relative acceleration to

$$q_{tt} = \frac{(-2|w|^2 + \rho h)q}{\rho^3} + P = -\frac{Mq}{|q|^3} + P.$$

The equations for R, U, y, V reconstruct the remaining Newtonian components. Deck equivariance follows by direct substitution. The implementation also contains an exact symbolic audit of these projection and gauge identities. $\square$

3 A posteriori tubes and arithmetic semantics

Let F denote either the ordinary field or the selected LC field, and let $\bar{X}$ be a polynomial on an interval whose maximum distance from its anchor is ℓ . Suppose outward interval evaluation proves, throughout a declared radius- r tube,

$$\|\bar{X}' - F(\bar{X})\|_\infty \leq \delta, \quad \|DF\|_\infty \leq K,$$

and the exact anchored state is within ε of the polynomial anchor. Set

$$E = e^{K\ell}\varepsilon + \delta \frac{e^{K\ell} - 1}{K}, \quad (7)$$

with $E = \varepsilon + \delta\ell$ when $K = 0$.

Lemma 3.1 (A posteriori tube). *If every denominator of F is separated from zero on the declared tube and $E < r$, then the exact anchored IVP has a unique solution on the complete chart interval and $\|X - \bar{X}\|_\infty \leq E$ there.*

Proof. Subtract the exact and polynomial integral equations. Grönwall's inequality yields (7) up to any first exit. At a hypothetical first exit the strict estimate gives $E < r$, a contradiction. Local uniqueness joins the anchored solution across the interval. $\square$

The ordinary checker uses an anisotropic version of this lemma. With position and velocity radii R_q, R_v , it evaluates the norm

$$\|(\Delta q, \Delta v)\|_W = \max\{\|\Delta q\|_\infty/R_q, \|\Delta v\|_\infty/R_v\}.$$

If A bounds the acceleration Jacobian, a valid Lipschitz bound is

$$L_W = \max\{R_v/R_q, AR_q/R_v\}.$$

The corresponding weighted defects and anchor errors are inserted in Lemma 3.1. This avoids forcing the position and velocity errors to have the same scale near a binary encounter.

3.1 Exact binary64-dyadic inputs

The raw version-1 loader accepts finite built-in binary64 values in real numeric fields; schema versions and counts are strict built-in integers. In every theorem-facing rational calculation, a binary64 value is interpreted as the exact dyadic rational represented by its bit pattern. Thus the token `0.1` denotes the exact round-tripped binary64 number, not the mathematical rational $1/10$. Exact equality means equality of the corresponding rational values. This convention prevents a certificate from silently changing the serialized IVP to a nearby decimal problem.

Exact integer and `Fraction` arithmetic is used for polynomial endpoints and handoff translations. Other algebraic and transcendental bounds required by the tube checkers use the directed outward operations described in Section 10.

3.2 Version-2 outward mass coefficients

Generic binary64 masses need not have binary64-representable derived ratios. The new theorem-facing LC tube, carried entry, and carried exit therefore do not compare a rounded ratio with an exact ratio. They first convert m_i, m_j, m_k to exact fractions and derive exactly

$$M, \quad \frac{m_j}{M}, \quad \frac{m_i}{M}, \quad \frac{m_k}{M}, \quad \frac{m_i m_k}{M}, \quad \frac{m_j m_k}{M}, \quad m_i \left(1 + \frac{m_k}{M}\right), \quad m_j \left(1 + \frac{m_k}{M}\right). \quad (8)$$

Each exact coefficient is then converted to the tight enclosing binary64 interval before it is consumed by the interval-dual LC field, its Jacobian, or the complete projection formulas.

Proposition 3.2 (Mass-system fidelity). *Subject to sound directed conversion and interval arithmetic, every accepted version-2 LC tube, carried entry, and carried exit encloses the exact LC field and projection for the exact binary64-dyadic mass triple serialized at the root.*

Proof. All coefficients in (8) are rational functions of positive exact rational masses, so their exact values are defined. Inclusion isotonic evaluation with an outward interval containing each value encloses the same exact-mass field and projection. No nearby rounded mass tuple is substituted. $\square$

An obligation may remain unresolved if an outward coefficient is not finite or if a later interval operation overflows. That is checker incompleteness, not a statement about the physical solution. This migration is intentionally limited to the v2 LC-tube/carried-entry/carried-exit/finite-chain surface; legacy LC checking paths retain their documented arithmetic gates.

4 The raw finite-chain certificate

The wire schema is `RawPlanarChainCertificate` version 1. The current top-level replay checker is version 2. The root contains positive masses, three initially distinct planar positions, three velocities, an initial physical time, and an ordinary-chart parameter, together with one ordinary chart and its tube. Acceptance requires zero binding tolerances and exact agreement among the binding parameter, the initial chart's left endpoint, and the tube anchor. The initial chart and tube are freshly replayed. The induction therefore starts from one exact point IVP rather than an initial-condition box.

The strict tagged-union grammar is

$$N(N \rightarrow N \mid N \rightarrow LC_{ij} \rightarrow N)^*, \quad (i, j) \in \{(0, 1), (0, 2), (1, 2)\}. \quad (9)$$

An `ordinary_bridge_v1` record contains a raw ordinary transition, target ordinary chart, and target tube. A `planar_lc_passage_v1` record contains a raw ordinary-to-LC entry, LC chart and tube, LC-to-ordinary exit, target ordinary chart, and target tube. Every LC segment returns to N before the next segment begins. Consequently the three pair charts can occur in any finite order, including revisits, without a direct transition between different LC coordinate systems.

Parsing is fail-closed. Exact classes, field sets, tags, canonical pair labels, round trips, and a globally unique identifier namespace are required. The raw wire does not contain a later IVP binding, a propagated clock origin, a selected gauge assignment, a projected endpoint, a cached checker result, or a final enclosure. Those objects are derived during replay. Canonical serialization and its aggregate SHA-256 bind transport and mutation of the raw record; the digest is not a substitute for any mathematical obligation.

5 The conditional induction kernel

After the root and after every fully accepted segment, the current vertex is an ordinary chart. The proof carries the following invariant.

Proposition 5.1 (Carried-branch invariant). *At a certified ordinary vertex there is one actual branch of the unique solution of the exact root IVP inside the freshly checked current tube. Its ordinary clock is $t = s + b$ for one actual scalar b in a checker-derived exact rational interval $B = [B_-, B_+]$. All accepted handoffs identify this branch with the same physical solution, modulo a pair-local LC deck transformation.*

This statement is existential in both the branch and the clock. It does not say that every state in the tube is on the root trajectory, that every $b \in B$ is realizable, or that arbitrary combinations

of state and clock interval coordinates are correlated. Complete interval containment is used only to prove that the one actual carried state remains included.

At the root, if the serialized initial physical time is t_0 and the initial ordinary parameter is s_0 , the invariant holds with the exact point

$$B_0 = [t_0 - s_0, t_0 - s_0]. \quad (10)$$

5.1 Ordinary bridge

Let e be the source ordinary chart's right parameter and a the target ordinary chart's left parameter. The bridge helper freshly checks both tubes, evaluates every source polynomial over the complete endpoint by exact rational arithmetic, inflates by the checked source error, and requires the resulting twelve-dimensional box to lie inside the target initial ball. Center or sampled agreement is insufficient.

For the actual carried branch, ordinary autonomy and uniqueness then identify the target solution with the source solution. Since the physical endpoint time is $e + b$, the target clock origin is $b' = e + b - a$. The interval update is derived exactly:

$$B' = B + e - a. \quad (11)$$

The cocycle record stores both exact intervals and the exact parameter translation; no producer-supplied clock is accepted.

5.2 Complete ordinary-to-LC entry

At the ordinary right endpoint the entry helper reconstructs the complete source box and requires the selected pair to be separated throughout it. A deterministic square-root grammar produces a complete finite LC lift cover. Depending on its relation to the square-root branch cut, that cover contains one or two patches. In the two-patch case their overlap determines an exact parity-one relation under (5). The checker constructs the exact $\mathbb{F}_2$ graph, enumerates its two global complements, and accepts only if one coherent complement places every complete emitted patch box inside one LC target anchor.

If D_{in} denotes the interval of entry physical times, it is derived from the source ledger as

$$D_{\text{in}} = B + e. \quad (12)$$

For the one actual carried source state, patch coverage supplies at least one exact constrained LC lift. The accepted coherent gauge sends that lift into the checked LC anchor, and its physical-time coordinate lies in D_{in} . The claim is not that every point of every rectangular patch box satisfies the nonlinear pair-energy constraint.

5.3 LC tube and strict physical time

The v2 LC checker freshly validates the entire fourteen-dimensional tube, including residual, derivative, Grönwall, third-body separation, and mass arithmetic obligations. The carried entry satisfies (6), Lemma 2.1 preserves it, and the punctured projection is the same exact-mass Newtonian branch.

The entry is separated, so $\rho > 0$ there and the analytic function z is not identically zero. Its zeros are therefore isolated. Since

$$\frac{dt}{ds} = |z|^2 \geq 0,$$

the physical-time increment over every nondegenerate LC parameter interval is strictly positive. This proves correct time ordering even if an isolated zero occurs. It does not prove that z has a zero.

5.4 Complete LC-to-ordinary exit

At the exact LC right parameter the exit helper evaluates all fourteen components and inflates them by the freshly checked LC error. It requires a strict positive lower bound for ρ on the complete exit slice. The full slice is then projected with the outward mass coefficients to twelve Cartesian position and velocity intervals, all of which must be contained in the target ordinary initial ball.

The LC physical-time component gives an outward interval

$$D_{\text{out}} = [d_-, d_+] \ni t_{\text{exit}}.$$

For target ordinary anchor a , the checker derives

$$B' = D_{\text{out}} - a. \quad (13)$$

Constraint invariance, strict LC time, deck-equivariant projection, complete endpoint containment, and ordinary uniqueness reestablish Proposition 5.1 at the target vertex.

5.5 Local gauges and repeated pairs

Each LC passage has its own patch namespace and one coherent gauge assignment. Gauges belonging to different selected pairs are never compared. Two visits to the same pair separated by an ordinary chart are also independent lift problems; the checker does not impose an unsupported equality between their gauge bits. Ordinary Cartesian coordinates are the common physical representation through which pair changes and revisits compose.

6 Fixed-time soundness theorem

After all supplied segments are accepted, let the current ordinary chart have domain $[a, r]$, and let the carried clock origin satisfy $b \in B = [B_-, B_+]$. For a finite requested physical target T , the checker first requires

$$T \geq a + B_+. \quad (14)$$

It then forms the exact parameter preimage interval

$$J = T - B = [T - B_+, T - B_-] \quad (15)$$

and requires $J \subseteq [a, r]$. Each final position and velocity polynomial is evaluated on all of J by exact `RationalInterval` Horner arithmetic. The current ordinary tube is replayed and its validated error is added outward. Exact rational endpoint differences determine the largest component width, which must not exceed the requested bound.

Theorem 6.1 (Raw finite planar continuation-chain soundness). *Let C be a canonically constructed raw version-1 certificate whose root is one exact positive-mass planar point IVP with initially distinct positions, and let T be its finite target. Subject to the trusted kernel of Section 10, if version-2 replay returns*

$$\text{check}(C).\text{status} = \text{CERTIFIED_TO_}T,$$

then:

- (i) *the exact root IVP has one unique chain-compatible generalized planar continuation through the supplied finite chain to physical time T ;*
- (ii) *its ordinary pieces solve (1), and every punctured constrained LC projection is that same Newtonian solution;*

- (iii) every handoff carries the same actual branch at one globally coherent physical clock, with strictly forward time through each LC passage;
- (iv) all selected pair changes and repeated visits compose through ordinary coordinates with only pair-local deck ambiguity;
- (v) the returned ordinary box contains the state at exactly T ; and
- (vi) every returned component width satisfies the requested bound.

Uniqueness here is relative to the supplied chain and its selected regularized lift; it is not a classification of every possible generalized continuation at a collision.

Proof. The exact root binding and fresh ordinary tube establish Proposition 5.1 with (10). Induct on the finite segment tuple. A complete ordinary bridge preserves the actual branch and derives (11). A complete LC passage covers the actual ordinary endpoint by a constrained lift, selects one coherent local gauge, validates its unique lifted evolution, projects the full separated exit into the next ordinary anchor, and derives (13). Thus every accepted segment reestablishes the invariant for the same root solution. LC strict-clock monotonicity forbids time reversal. At the terminal vertex, the actual $b \in B$ implies $s_T = T - b \in J$. Exact Horner evaluation on all of J , followed by the fresh tube inflation, contains the state at T . The exact width gate proves the last conclusion. $\square$

Remark 6.2 (No event conclusion from an ordinary LC passage). *Theorem 6.1 permits regularized coordinate passages. It does not assert that $z = 0$, or equivalently $\rho = 0$, occurs in any such passage. Collision occurrence, isolation, and time require a separate checked event witness.*

7 Unresolved frontiers and replay semantics

Malformed canonical JSON, unknown fields or tags, wrong exact classes, and other parser violations may raise before any replay result exists. For a successfully constructed exact-class certificate, a failed tube, containment, arithmetic, target-domain, or width obligation returns UNRESOLVED. It is not a claim of nonexistence, blow-up, or collision.

Replay counts the longest consecutive segment prefix accepted before the first failed segment and retains at most one typed region:

- the current ordinary right frontier after a certified root or prefix;
- a fourteen-dimensional LC right frontier when entry and LC-tube replay succeed but the failing passage's exit does not; or
- the ordinary fixed- T enclosure when mathematical enclosure succeeds but the requested width gate fails.

No certified root means no certified prefix and no retained region. An LC frontier is never relabeled as a finite Cartesian velocity state. If a frontier time interval contains the unknown actual endpoint, only its lower endpoint is reported as guaranteed reached. An unresolved result proves only the accepted prefix and its typed containment statement; it says nothing about continuation or certifiability beyond that frontier.

The result retains the raw certificate, its canonical aggregate digest, derived ledgers, checker and kernel identifiers, and the obligation ledger. Its certification property reruns the same top-level implementation and requires exact result equality. Mutated evidence, hostile subclasses, copied all-true ledgers, changed identifiers, and altered derived snapshots do not inherit certification. This is exact-self replay, not an independently implemented verifier or a proof-assistant proof.

For review, a successful top-level result pins the identifiers `raw_planar_continuation_chain_replay_checker_v2`, `exact_root_and_conditional_planar_segment_induction_kernel_v1`, `forward_interval_clock_origin_ledger_v1`, and `exact_rational_horner_fixed_time_kernel_v1`. Its segment ledger admits only `carried_ordinary_bridge_checker_v1` and `carried_planar_lc_exit_checker_v2`. The latter self-binds `carried_planar_lc_entry_checker_v2`, `independent_planar_lc_aposteriori_tube_checker_v2`, and the mass kernel `planar_lc_mass_coefficients_exact_binary64_fraction_outward_v1`. These strings are implementation identity pins, not evidence that the so-named components are independent implementations of the top-level replay.

8 A separate event-anchored collision theorem

The repository retains a stronger local checker for a supplied LC chart whose anchor is asserted to be the collision event. It is logically separate from the generic finite-chain passage.

Theorem 8.1 (Exact event-anchored binary-collision passage). *Fix positive masses and one selected pair. Suppose the event checker accepts an LC tube on $[s_-, s_+]$, $s_- < 0 < s_+$, with zero anchor error at $s = 0$, and checks exactly*

$$z(0) = 0, \quad 2|w(0)|^2 = M. \quad (16)$$

Suppose it also validates separated-third-body denominators on the complete LC tube, positive ρ on both complete punctured endpoint slices, complete LC-to-Newton projection there, and containment in validated ordinary tubes. Then the exact lifted IVP has an isolated selected-pair collision at $s = 0$; its two punctured projections are branches of the same Newtonian trajectory, and physical time is strictly ordered from the left endpoint through the collision to the right endpoint.

Proof. Lemma 3.1 gives one analytic lifted solution. The exact anchor sets $C = 0$ and, because $M > 0$, gives $w(0) \neq 0$. Hence z has a simple isolated zero. In particular,

$$t(s) - t(0) = \int_0^s |z(u)|^2 du = \frac{|w(0)|^2}{3} s^3 + O(s^4).$$

Lemma 2.1 supplies the Newtonian equations off the event. Projection of the complete endpoint slices and ordinary uniqueness bind both punctured branches to the same lifted IVP. Finally, $t' = |z|^2$ and nonidentity of analytic z imply strict physical-time ordering. $\square$

The zero-error constraint anchor is essential: a positive box centered on a constrained point generally includes off-constraint states, whose projected singular coefficient is not the same Newtonian coefficient. This theorem is therefore an event theorem only when the exact identities (16) are checked. The retained event checker has its own documented mass-consistency restrictions; Proposition 3.2 does not claim that every legacy collision or overlap path has been migrated to the v2 outward mass kernel.

9 Review artifact and reproducibility

The primary review evidence is raw JSON, not the candidate builder. The script

`scripts/certify_repeated_planar_chain.py`

constructs candidate fixtures, exports canonical raw records, performs fresh replay, and verifies a transport manifest. Its candidate-construction path is untrusted: only the raw record and the obligations recomputed from that record enter Theorem 6.1.

The version-0.3 review bundle is located at

artifacts/v0.3.0-review/planar-chain/

and is organized around the following files:

- `success.raw.json` and `success.replay.json`;
- `failed-revisit.raw.json` and `failed-revisit.replay.json`; and
- `manifest.json`.

The success fixture has the five-segment chart word

$$N \rightarrow N \rightarrow LC_{01} \rightarrow N \rightarrow LC_{02} \rightarrow N \rightarrow LC_{12} \rightarrow N \rightarrow LC_{01} \rightarrow N.$$

It exercises all canonical pairs and a same-pair revisit. The companion negative fixture is designed so that the final revisit exit fails after a four-segment certified prefix, retaining a typed fourteen-dimensional LC frontier. These expected fixture outcomes are regression evidence for the finite grammar and failure semantics, not proof that any LC passage contains a physical collision.

For the candidate bytes selected for the review bundle, fresh same-implementation replay gave the following checked values. The success raw SHA-256 is `ede15b0f35cf741f542a6cd260470a93ee5ff85dc5ae2371db1b88819b821f11`; its status is `CERTIFIED_T0_T` after five segments. The requested target is approximately 0.008008225523335941, the exact final parameter preimage converts to approximately $[0.005, 0.009000009750858424]$, and the reported maximum component width is approximately $0.021640584283380553 < 0.1$. The negative raw SHA-256 is `c6830919726c22fbca6e45d5781b2465e54876bfa89c25bc26e97284ff918727`; its status is `UNRESOLVED`, with prefix count four, failed segment index four, first failing obligation `carried_lc_exit_target_initial_ball_contains_complete_projection`, and a retained fourteen-dimensional LC frontier.

The canonical manifest remains the source of the machine-readable values. Its hashes bind transport bytes, not mathematical truth. Environment versions are informational and are not silently promoted to theorem assumptions. Fresh-process export comparison and bundle verification are used to detect nondeterministic serialization. The replay transcripts are generated by the same Python implementation as the checker and must not be described as an independent verification.

The local projection algebra audit remains available through

`scripts/verify_lc_projection_identities.py`.

The repository documentation records focused and full regression commands. Release review should use the tracked raw bytes, rerun the checker in a clean environment, compare the new transcript with the tracked one, and report any difference rather than editing a result record by hand.

10 Trusted kernel and precise nonclaims

Theorem 6.1 is conditional on the following trusted kernel:

- Python integer and `Fraction` arithmetic, strict class checks, canonical JSON handling, and SHA-256 transport hashing;
- exact `RationalInterval` polynomial arithmetic and binary64-to-exact-fraction conversion;
- directed `nextafter` inflation and the scalar outward bounds used for square roots, reciprocal powers, exponentials, Grönwall estimates, square-root lifts, and projections;

- interval forward automatic differentiation for the ordinary and LC fields, including complete Jacobian row-sum bounds;
- the explicit Newtonian and LC vector fields, mass-coefficient formulas, constraint, lift, projection, and deck action;
- the a posteriori ODE lemma, analytic local existence and uniqueness, constraint invariance, punctured projection equivalence, deck equivariance, and the nontrivial-analytic- z strict-clock lemma;
- complete interval containment, the canonical square-root cover, the exact $\mathbb{F}_2$ graph kernel, conditional segment induction, the forward clock ledger, and fixed-time evaluation; and
- correctness of the Python runtime and of the implementation of these kernels.

IEEE 1788 provides historical standardization context for interval arithmetic [2]; a general reference is Moore, Kearfott, and Cloud [6]. In the present binary64 backend, sensitive quantities are inflated outward and some scalar bounds use directed high-precision decimal arithmetic. In particular, `Decimal.exp` under its directed context is part of the runtime hypothesis; this manuscript does not prove it to be a correctly rounded transcendental implementation. Adversarial tests support the arithmetic kernel but do not replace its audit.

The current result does *not* establish any of the following:

- a complete or terminating certificate producer for arbitrary initial data;
- a theorem for boxes or families of IVPs;
- that every point in a state or clock interval is realizable;
- automatic collision detection, collision occurrence in an ordinary LC passage, or collision isolation without a separate event witness;
- continuation through simultaneous or total collision;
- spatial motion, Kustaanheimo–Stiefel chart gluing, or a spatial finite-chain theorem;
- escape completeness, non-Zeno progress, all-time coverage, or an infinite-atlas theorem;
- an elementary expression, closed form, or general closed solution of the three-body problem;
- migration of every legacy LC checker to outward mass arithmetic;
- an independently implemented verifier, an arbitrary-precision audited backend, or a proof-assistant formalization;
- independent external mathematical review or institutional endorsement; or
- correctness without the arithmetic, analytic, runtime, and implementation assumptions above.

Sampled residuals, nominal physical-time metadata, producer success flags, reason strings, producer-selected gauge bits, object subclasses, and cached checker results are not theorem hypotheses. The scientifically defensible claim is narrower: accepted raw evidence carries one supplied exact planar point IVP through one supplied finite ordinary/LC chain to a rigorous fixed-time enclosure.

11 Directions toward broader certified computation

The present theorem supplies the soundness half of a proof-carrying integrator: candidate construction may be heuristic, but every accepted step is checked from raw evidence and composes through an explicit invariant. A broader computational result now depends less on adding chart types to the statement and more on proving progress properties for a producer. Concrete next steps are:

- (i) build adaptive ordinary and LC certificate producers whose output is measured by acceptance rate, certified time advance, and enclosure growth rather than nominal numerical success;
- (ii) add rigorous event isolation when a collision conclusion is required, instead of inferring an event from LC chart use;
- (iii) construct an independent verifier, preferably with an audited arbitrary-precision interval backend, and compare canonical raw transcripts across implementations;
- (iv) formalize the finite induction and clock ledger in a proof assistant, leaving only explicit arithmetic certificates to an executable kernel;
- (v) study validated families and parameter boxes, beginning with robust noncollision regimes and isolated binary-encounter scattering maps; and
- (vi) formulate and test quantitative progress hypotheses needed for escape coverage, collision classification, and exclusion of Zeno switching.

None of these steps is required for the finite acceptance theorem, and none is claimed by the current artifact. Together they give a falsifiable path from one proof-carrying finite trajectory to a genuinely broader certified computational solution.

12 Software availability

The implementation is available under the BSD-3-Clause license at

<https://github.com/jspaulding42/three-body-regularized-atlas>.

The immutable tag `v0.2.0-review` contains the earlier local collision-passage snapshot. This manuscript describes the prospective `v0.3.0-review` finite-chain artifact; no `v0.3` tag or archival DOI is claimed until the release is actually published. The repository contains the raw schema, replay implementation, candidate builder, tests, review bundle, and the exact symbolic projection audit.

Generative-AI use and contributions

This project was conducted through human-directed use of OpenAI generative-AI systems identified in the user interface as GPT-5.5, GPT-5.5 Pro, and GPT-5.6. Those systems contributed substantially to mathematical exploration, software implementation, certificate and test design, manuscript drafting, and critical revision. Jeffrey Spaulding initiated and directed the project, selected its research direction and scope, reviewed and authorized the release of the resulting artifact, and is the sole accountable author. The AI systems are not authors: they cannot approve the manuscript, hold rights, disclose conflicts, or accept responsibility for its contents. The manuscript and software should therefore be read as a human-released, AI-assisted research artifact whose mathematical claims remain subject to independent expert review. No institutional endorsement or independent external mathematical review is asserted.

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